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Forecasting on PM10 in South Korea Using the CNN-BiGRU-LSTM Model

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ABSTRACT

After the COVID-19 pandemic in Korea and worldwide, factory operations have resumed. As environmental pollution accelerates, the number of days with a severe fine-dust concentration has been rapidly increasing. The higher fine-dust concentration results in more dust entering the lungs or brain, adversely affecting the body. This study was conducted to guide national or personal alternative methods of prevention when the fine-dust concentration is severe. If the fine-dust concentration forecasting system is accurate, high concentrations could be more reliably prevented. Based on the weather and air pollution variables, this study focused on improving predictive performance by comparing the MLP, basic ANN models, RNNs (RNN, LSTM, and GRU), and CNN, which are time-series DL models, and implementing a new hybrid combination model. As a result, the hybrid models, CNN-BiGRU and CNN-BiGRU-LSTM, improve prediction performance better than the single DL models.

1 | Introduction

Fine dust refers to particles too small to see and are classified by particle size. Fine dust is particulate matter (PM) floating in the atmosphere, where PM_{10} dust has a diameter of $10\,\mu m$ or less and $PM_{2.5}$ is $2.5\,\mu m$ or less. The size of PM_{10} is about one-fifth to one-seventh of the diameter of a strand of hair, and $PM_{2.5}$ is so small that it is only about one-twentieth to one-thirtieth of a strand of hair. The World Health Organization (WHO) announced that 7 million people died earlier than expected due to fine dust in 2014. In 2013, the International Agency for Research on Cancer under WHO classified fine dust as a Group 1 carcinogen. Domestic news in Korea indicates that fine dust frequently occurs, and public interest has rapidly increased. According to data from the Organisation for Economic Co-operation and Development (OECD), Korea's fine-dust concentration in 2020 was $25.3\,\mu g/m^3$, the highest among comparative countries. That amount is about twice as high as the OECD average ($13\,\mu g/m^3$) and more than four times as severe as Finland ($5.6\,\mu g/m^3$). In addition, in 2023, the average concentration of ultrafine dust in Korea rebounded for the first time in 7 years to $18.3\,\mu g/m^3$. Last year, after Korea and other countries worldwide declared the coronavirus disease 2019 (COVID-19) pandemic, factories resumed operation, and experts analyzed that the changes in the weather environment are a primary cause of the increased fine-dust concentration [1]. The environmental standards for fine dust in Korea have been in effect since 2015. In 2015, the government announced it would set standards and develop a forecast system by designating fine dust as an air environment reference material [2]. In addition, since 2013, the media and WHO have stated that air pollution causes diseases on par with unhealthy diets and smoking. It causes ischemic heart disease and stroke in adults, and children suffer from reduced lung function and respiratory diseases, causing 7 million premature deaths every year. If air pollution is reduced to the level proposed in the guidelines, about 80% of ultrafine dust-related deaths can be avoided. However, fine dust is associated with

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numerous factors, such as industrial processes, and it is impossible to remove it altogether. Therefore, an accurate fine-dust concentration forecasting system would allow the preparation of alternative methods. Thus, studies on improving the performance of predicting fine-dust concentrations have been conducted at home and abroad [3]. With the development of artificial intelligence algorithms, research has continued on machine learning and deep learning (DL) prediction models. In the past, a single time-series model was applied to predict the concentration of fine dust [4]. As these models were developed, studies were conducted to compare their predictive performance. Fine-dust concentration data and variable data were used, such as concentrations of sulfur dioxide (SO_2), PM_{10}, nitrogen dioxide (NO_2), nitrogen oxide (NOx), carbon monoxide (CO), ozone (O_3), benzene (C6H6), and other air pollutant data using various models, such as support vector regression, random forest, k-nearest neighbors, gradient boosting regression, artificial neural networks (ANNs), and recurrent neural networks (RNNs). Single DL models, such as the convolutional neural network (CNN), gated recurrent units (GRU), and long short-term memory (LSTM), were the most frequently applied, followed by research cases using statistical models, such as Bayesian model averaging [5–11]. Subsequent studies have explored optimization techniques to improve the performance of each model [12]. In addition, fine-dust concentration prediction can be improved using additional parameters. For example, Ramli et al. [13] predicted the next day's fine-dust concentration in the Malaysian Peninsula using the Bayesian model averaging method with eight air-quality monitoring data parameters. This model evaluates a combination of models as a statistical model-fit method considering uncertainty, confirming the best predictive performance when relative humidity, wind speed, and the model were applied [13]. These studies consider additional air-quality parameters while using weather variables, such as temperature, humidity, wind speed, and wind direction, or compare predictive performance by combining variables. Fernando et al. [14] aimed to predict PM_{10} using air pollution data and weekend/weekday data, which are temporal factors. Recently, hybrid models combining basic time-series models, ANNs, and DL models have been developed [15–17]. For example, Alyousifi et al. [18] proposed a hybrid fuzzy logic-based model incorporating DL and meteorological variables and showed that this approach is effective in improving air-quality prediction performance in urban areas. In a related study, Bitencourt et al. [19] developed a hybrid model combining fuzzy logic with Deep Belief Networks (DBN) for air pollution prediction. Their approach demonstrated higher accuracy and robustness compared to traditional machine learning models, highlighting the effectiveness of fuzzy logic in handling uncertainty in input variables and improving model stability. These are about hybrid fuzzy logic techniques. And Adnane et al. [20] developed the nonlinear autoregressive natural network with multiple exogenous variables, and Isikdag [21] confirmed that the nonlinear autoregressive natural network model revealed the most accurate prediction results in the prediction of the fine-dust concentration time series with short-term time dependence [20, 21]. Veleva et al. [22] applied the Box-Jenkins autoregressive integrated moving average model to the fine-dust concentration measurement data collected by Bulgaria's Ministry of Water Resources from 2010 to 2021 and combined the generalized autoregressive conditional heteroskedasticity model to consider the variability of the residuals. The model effectively predicts the variability of the annual fine-dust concentration. Wang et al. [23] proposed a residual analysis and regression-based autoregressive and moving average model combining classical time-series analysis methods and regression analysis. In addition, research has been conducted to apply the clustering method to take advantage of regional characteristics. Francisi et al. [24] used the k-means clustering method and a model combining the ANN and multilayer perceptron (MLP), arguing that the clustering method in highly polluted areas is effective. In Korea, studies have been conducted to predict fine-dust concentration in domestic regions using single machine learning models [25–29]. In addition, for fine dust generated in a specific area, applying pollutant data from factories to predict the fine-dust concentration in an industrial complex would be advantageous [30]. While previous studies have largely focused on single DL architectures or short-term forecasting, the application of a triple-hybrid structure combining CNN, BiGRU, and LSTM for long-term PM10 forecasting remains limited, especially within the context of South Korea. In this study, long-term forecasting refers to predicting PM10 levels 24 h or more into the future. This study attempts to fill that gap by implementing and evaluating this structure on real urban datasets. Therefore, this study compares the predictive performance of machine learning models suitable for time-series data and CNN models, which are commonly used for image classification but are also applied to prediction, and proposes a hybrid model based on the CNN (with relatively good predictive performance [31]). Section 2 describes the CNN-bidirectional GRU (BiGRU)-LSTM model combining the MLP, RNN, LSTM, GRU, and CNN models to predict the fine-dust concentration and the time-series DL models based on the CNN. Section 3 explains the weather and air pollution data and preprocessing methods, and Section 4 compares the predictive performance after applying the machine learning models in Section 2. Finally, Section 5 discusses the conclusions and future research directions.

2 | Methodology

This section introduces the methods used to predict the fine-dust concentration. First, this study applies the basic ANN and a single DL model. This section describes a hybrid model combining a DL model based on the CNN.

2.1 | Multilayer Perceptron (MLP) Neural Network Model

The MLP neural network model is well known and widely used. On most occasions, the signals are transmitted in the network in one direction: from the input layer to the output layer [31]. This

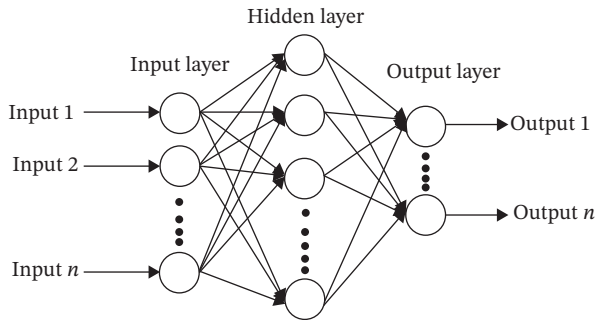

FIGURE 1 | Multilayer perceptron with two hidden layers.

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model has input, hidden, and output layers, as illustrated in Figure 1.

The input layer accepts the data to process and transfers them to the next step. Afterward, the hidden layers process the data input from the previous layer via activation functions and weights. The output layer provides the predicted data based on the processed data. Weighing is adjusted so that the value output is as close to the actual value as possible [31].

2.2 | Recurrent Neural Network (RNN) Model

The RNN has connections between the computational units from a directed circle. Unlike feed-forward networks, the RNN can use internal memory to process an arbitrary sequence of inputs. The RNN is created by applying the same set of weights recursively over a graph-like structure [32]. The RNN updates the hidden state h_t for a given sequence $x = (x_1, x_2, \cdots, x_t$ using the following equation:

$$h_{t-1} = \begin{cases} 0, & t = 0 \\ \phi(h_t, x_t), & \text{otherwise} \end{cases} \tag{1}$$

where *phi* represents a nonlinear function. The RNN outputs a probability distribution for the next element of the sequence, considering the hidden state h_t, and the probability of such a sequence can be decomposed in the following equation:

$$p(x_1, \cdots, x_t) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2) \cdots p(x_t|x_1, \cdots, x_{t-1}). \tag{2}$$

Each conditional probability distribution is expressed as follows:

$$p(x_t|x_1, \cdots, x_{t-1}) = g(h_t). \tag{3}$$

The above equation reflects the fact that the x_t in the sequence depends on the elements and changes due to the previous elements, allowing it to learn and predict complex patterns [33].

2.3 | LSTM Model

The LSTM is an RNN designed to solve the problem of the previous information in long-sequence data not affecting the information gained later. The LSTM proceeds through memory cells and three types of gates: input, forget, and output gates. Through this process, the dependence relationship is remembered between sequence data. Each gate is represented by the following equations [34]:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i), \tag{4}$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), \tag{5}$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o). \tag{6}$$

In the above equations, for the cell state, the input gate i_t determines the stored information, the forget gate f_t determines the deleted information, and the output gate o_t determines the output information. Moreover, h_t denotes the hidden state, W and U indicate the weight for each layer, and b represents the deflection value for each layer. The gates are adjusted to values between 0 and 1 using the sigmoid activation function (σ) [35].

2.4 | Gated Recurrent Unit (GRU) Model

The GRU is a modified RNN that can process characteristics in which previous information affects later data in a sequence of data while reducing the complexity of LSTM. The GRU model has only the update and reset gates, and it updates the gates by reducing the three gates of the LSTM to two. These two gates are critical in the progression of GRUs. The GRUs with these structures have fewer parameters than LSTM, resulting in a higher computational efficiency in the learning process. Nevertheless, in several studies, GRU primarily performs as well as or better than LSTM. The lower complexity due to reducing the number of gates did not significantly affect the performance of the GRUs [36].

2.5 | CNN Model

The CNN is an effective DL model for recognizing data patterns, such as images, preserving location information in data (i.e., the association and structure between pixels in the image), and extracting features via convolutional operations. Specifically, the convolutional layers apply the convolutional operation between the raw input data and convolutional kernels, producing new feature values. The input data must have a structured matrix form because this technique was initially intended for extracting features from image datasets [37]. The left side of Figure 2 illustrates the input time-series data, comprising a multidimensional matrix convoluted from top to bottom, as demonstrated by the arrow, and the red outlines represent a filter. The number of the extracted feature dimensions is $N \times 1$ after the convolution with a filter, where N is related to the number of input data dimensions, the size of the filter, and the convolutional step length. The blue outlines indicate another filter, which can be followed by further

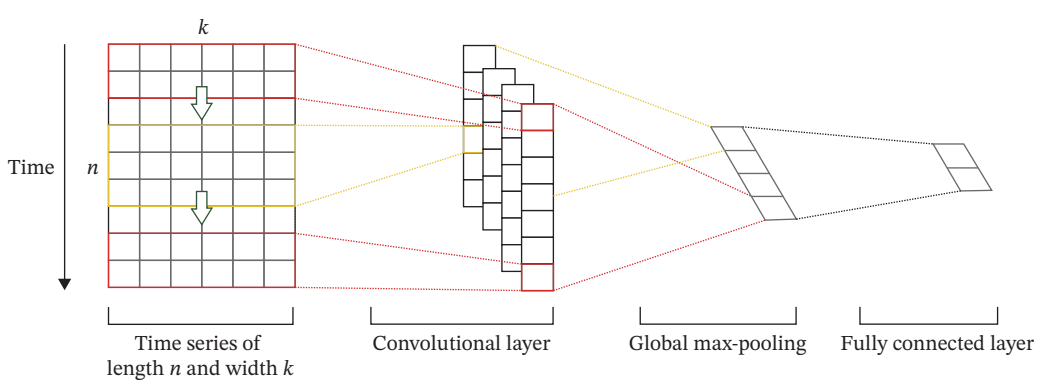

FIGURE 2 | One-dimensional convolutional neural network process.

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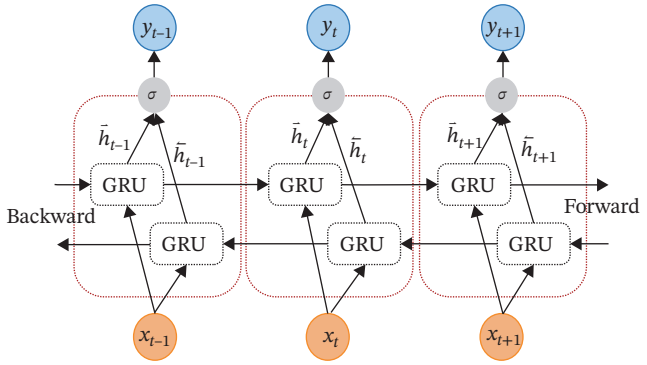

FIGURE 3 | Schematic of the bidirectional gated recurrent unit (BiGRU) model.

filters. If the number of filters is M, the extracted feature dimensions are $N \times M$ [38]. The CNN consists of several layers, including convolutional, pooling, and fully connected layers. The CNN learning process is conducted using a backpropagation algorithm, and the weights are updated to minimize the loss function [39].

2.6 | CNN-Bidirectional Gated Recurrent Unit (CNN-BiGRU) Model

The CNN-BiGRU model is improved using the objective hybrid DL model: CNN-GRU. The framework involves integrating the 1D CNN and BiGRU. The CNN model acts as a spatial feature extractor, whereas the newly generated GRU computationally efficiently forecasts air pollutants for time-series prediction [8]. The BiGRU network can learn the relationship between past and future load-influencing factors and the current load, which is more conducive to extracting the deep features of load data [40]. Figure 3 depicts the structure of BiGRU.

2.7 | Hybrid CNN-Bidirectional Gated Recurrent Unit-LSTM (CNN-BiGRU-LSTM) Model

Figure 4 describes the hybrid CNN-BiGRU-LSTM model in detail. This model has two central parts, feature extraction and prediction, each with several layers. Feature extraction is a one-layer CNN, and its input is the weather conditions (i.e., temperature and humidity) and air pollutants (i.e., CO_2, NO_2, etc.). In the prediction part, the BiGRU-LSTM model outputs the corresponding prediction values [30, 41]. Although the proposed model architecture is relatively complex, the average training time was within 3 h. To prevent overfitting, early stopping was applied. Furthermore, ~2.3% of the dataset contained missing values, which were imputed using linear interpolation methods.

TABLE 1 | Variable statistics by city.

State PM_{10}	Dongducheon-si	Dongjak-gu	Jung-gu
Mean	40.12	42.56	40.39
SD	33.67	31.44	34.07
Min	1	3	0
Q1	19	24	21
Median	32	36	33
Q3	52	53	50

3 | Data Processing and Results

3.1 | Data Preprocessing

The data used in this study are publicly available from the Korea Environment Corporation (https://www.airkorea.or.kr) and the Korea Meteorological Administration (https://data.kma.go.kr). All data are open access and can be obtained freely from the corresponding websites. For each city, the mean PM10 concentration was calculated and used as representative data for the analysis. While peak prediction remains a challenge for many DL models, average-level forecasting still provides practical value. Accurate trend predictions can inform daily air-quality advisories, optimize public health messaging, and support the deployment of preventive actions such as air purifier operation or activity restrictions. Furthermore, real-time integration of peak-detection layers or ensemble models may improve the model's response to high-risk events in future work. The Korea Meteorological Administration collected the meteorological data. Table 1 lists the data from

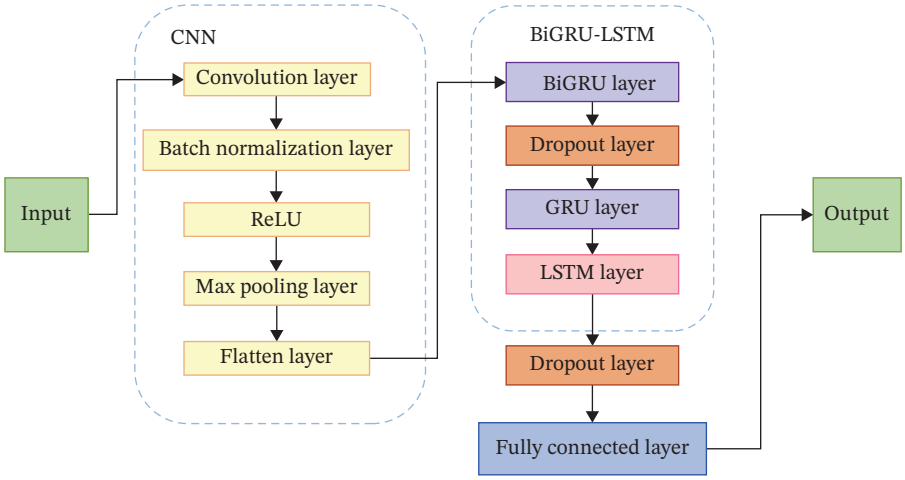

FIGURE 4 | Structure of the hybrid convolutional neural network (CNN) bidirectional gated recurrent unit (BiGRU) long short-term memory (LSTM) model.

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TABLE 2 | Weather variable description, all measured for 1 h of observation.

Weather (label and units)	Description
Temperature (temp °C)	Temperature of the atmosphere measured at a height of 1.5 m above the ground because that is the average height of a person
Wind speed (ws m/s)	Wind speed; wind damage changes depending on wind speed
Wind direction (wd °)	Wind direction, continuously changing; average wind direction was taken for 10 min before the observation time
Humidity (humid %)	Amount of water vapor in the air; unlike fog, water vapor refers to gaseous water
Station pressure (hPa)	Pressure of the Earth's atmosphere at a certain point
SO_2 (ppm)	Colorless, noncombustible gas that smells unpleasant and is irritating a representative gaseous air pollutant of SOx
CO (ppm)	Colorless, odorless, and poisonous gas caused by incomplete combustion of fuel; high CO concentrations occur in urban areas
O_3 (ppm)	NOx and volatile organic compounds in the atmosphere generated via a catalytic reaction by ultraviolet rays are characterized by strong oxidation power; O_3 protects the ecosystem by blocking ultraviolet rays
NO_2 (ppm)	Reddish-brown toxic gas with an unpleasant smell; the most influential substance for air pollution and acts as a precursor for producing O_3
PM_{10} ($\mu g/m_3$)	Dust with a particle size of less than 10 μg

three cities, collected from January 1, 2017, to December 31, 2022. The dataset was divided chronologically, with data from January 2017 to December 2021 used for training and validation, and data from January to December 2022 used for testing. The three cities with the least missing weather and air pollution data were selected.

Temperature, precipitation, wind speed, wind direction, humidity, and local atmospheric pressure were the meteorological variables, and SO_2, CO, O_3, and NO_2 data were collected as air pollution variables. Table 2 describes the variables. Figure 5 presents the correlations between each meteorological variable, air pollution variable, and the dependent variable. The variables affecting each region differed.

For the precipitation, an extended period of missing data occurred and was supplemented using the linear interpolation method. However, the linear interpolation method must have data before and after; therefore, the section where long-term missing data occurred from January 1, 2017, was excluded from the data collection period. The correlation was confirmed between the weather and air pollution variables and the dependent variable, PM10. The effects of the variables differed from city to city, and in Jung-gu, Incheon, the effects of the weather variables were minimal compared to the other cities. The map of Jung-gu, Incheon, reveals that air pollutants may have a much more significant influence than weather variables (except for air pressure) because they are adjacent to the sea and to Incheon International Airport and driving centers, which generate considerable air pollution from airplanes and cars.

3.2 | Evaluation Metrics

The mean absolute error (MAE) and root mean squared error (RMSE) were applied to evaluate the predictive performance of each model. The mean absolute percentage error (MAPE) is widely used for model evaluation. However, if the fine-dust concentration prediction value is less than or close to 0, the MAPE value becomes extremely large or cannot be calculated. Therefore, accuracy was evaluated using the MAE and RMSE scales defined as follows:

$$\mathrm{MAE} = \frac{\sum_{t=1}^{n} |Y_t - F_t|}{n}, \tag{7}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{t=1}^{n} (Y_t - F_t)^2}{n}}, \tag{8}$$

where n denotes the number of data used for prediction, Y_t represents the value observed at a point in time, and F_t indicates the value predicted using the model at that point in time. Lower MAE and RMSE values indicate better accuracy.

3.3 | Model Performance Comparison

3.3.1 | Hyperparameter Optimization

GridSearchCV was applied to each single DL model for hyperparameter optimization. The values of n_{unit} commonly applied for each model are 16, 32, 64, 128, and 256, and the Adam and root mean square propagation (RMSProp) optimizers selected the hyperparameter with the smallest MAE value. Table 3 presents the results.

3.3.2 | Performance Results

Table 4 presents the model application results. The hyperparameters of the hybrid CNN-BiGRU and CNN-BiGRU-LSTM models were selected based on the best performance based on the MAE. The model demonstrating the best prediction performance was highlighted in bold. In previous studies, performance was compared using only the MLP, RNN, LSTM, and GRU models, and the CNN models were added to improve performance further. In addition, rolling predictions for the fine-dust concentration per hour were conducted, and the number of characteristics and past data values were not often used. Moreover, each input sample

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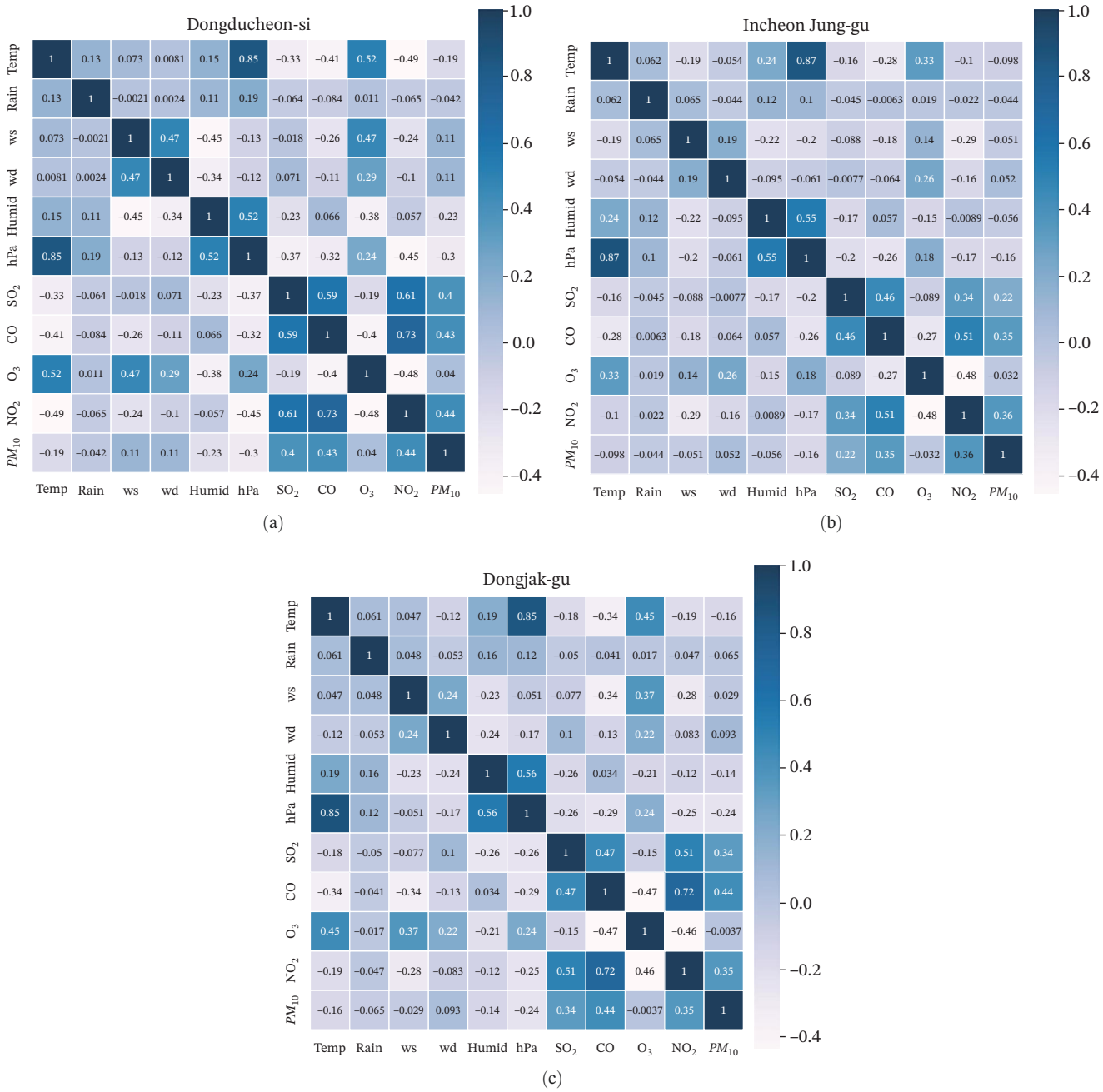

FIGURE 5 | Graph of the correlation between the fine-dust concentration and independent variables. (a) Dongducheon-si, (b) Jung-gu, and (c) Dongjak-du.

TABLE 3 | Optimization hyperparameter validation.

	MLP		RNN		LSTM		GRU		CNN	
Station	**n_unit**	**optimizer**	**n_unit**	**optimizer**	**n_unit**	**optimizer**	**n_unit**	**optimizer**	**n_unit**	**optimizer**
Dongducheon-si	16	Adam	32	RMSProp	16	RMSProp	128	RMSProp	64	RMSprop
Jung-gu	258	RMSProp	256	RMSProp	16	RMSProp	256	Adam	256	Adam
Dongjack-gu	128	Adam	64	Adam	64	Adam	128	RMSProp	64	RMSprop

includes time-lagged variables from the preceding 24 h, providing temporal context for the target PM10 concentration. Therefore, the CNN structure was applied as a 1D convolutional layer (Figure 6). A hybrid model was implemented by taking the feature values calculated from the CNN model and combining the LSTM model with the layer that added the bidirectional function to the GRU model that performed well in the previous single DL model [30]. Table 4 confirms that the CNN-BiGRU-LSTM model had the

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TABLE 4 | Summary of model results by city.

State	Models	MAE	RMSE
Dongducheon-si	MLP	10.81	20.80
	RNN	11.19	21.28
	LSTM	10.90	20.70
	GRU	11.22	21.25
	CNN	13.10	22.01
	CNN-BiGRU	10.55	20.96
	CNN-BiGRU-LSTM	**10.50**	**20.84**
Dongjak-gu	MLP	15.65	26.21
	RNN	14.03	24.52
	LSTM	13.22	23.72
	GRU	12.66	23.19
	CNN	13.00	23.45
	CNN-BiGRU	12.26	22.66
	CNN-BiGRU-LSTM	**12.17**	**22.40**
Jung-gu	MLP	9.92	17.46
	RNN	11.46	18.54
	LSTM	10.37	17.69
	GRU	9.93	17.61
	CNN	10.80	17.94
	CNN-BiGRU	10.21	17.96
	CNN-BiGRU-LSTM	**9.55**	**20.78**

Note: Bold values indicate the best performance for each category.

best performance, although it varies slightly from city to city. The next best-performing model was the CNN-BiGRU model. This result confirms that the hybrid model combining CNN and other DL models outperformed the predictive performance of the single CNN models and other DL models.

Figure 7 graphs the 2022 forecast value of the hybrid CNN-BiGRU model, which had the best predictive performance in Dongducheon-si, Gyeonggi-do. The red line represents the actual fine-dust concentration, and the light green line represents the fine-dust concentration predicted by the hybrid model. The fine-dust concentration value occasionally fluctuates. In Dongducheon, the CNN-BiGRU model performed better than the CNN-BiGRU-LSTM, possibly due to simpler pollutant patterns and the effective bidirectional sequence modeling of BiGRU without the added complexity of LSTM.

4 | Conclusion

After the COVID-19 pandemic in Korea and worldwide, factory operations have resumed. As environmental pollution accelerates, the number of days with a severe fine-dust concentration has been rapidly increasing. The higher fine-dust concentration results in more dust entering the lungs or brain, adversely affecting the body. This study was conducted to guide national or personal alternative methods of prevention when the fine-dust concentration is severe. If the fine-dust concentration forecasting system is accurate, high concentrations could be more reliably prevented. The fine-dust concentration (PM_{10}) was compared by applying models using eight weather and air pollution variables: temperature, wind speed, wind direction, air pressure, humidity, SO_2, CO, O_3, and NO_2. Initially, the optimal hyperparameters were determined and compared using GridSearchCV for a single RNN-based DL model (RNN, LSTM, and GRU). Additionally, the CNN, which performs predictions well in time-series models, has recently been added. Thus, the hybrid CNN-BiGRU and CNN-BiGRU-LSTM models that combine an RNN family model were implemented based on the CNN. In Dongducheon City, the CNN-BiGRU model had the best prediction performance with a value of 10.50 based on MAE, and the CNN-BiGRU-LSTM model had values of 12.17 and 9.55 in Jung-gu and Dongjak-gu, respectively. Therefore, the hybrid model improves prediction performance better than the single models. Based on the weather and air pollution variables, this study focused on

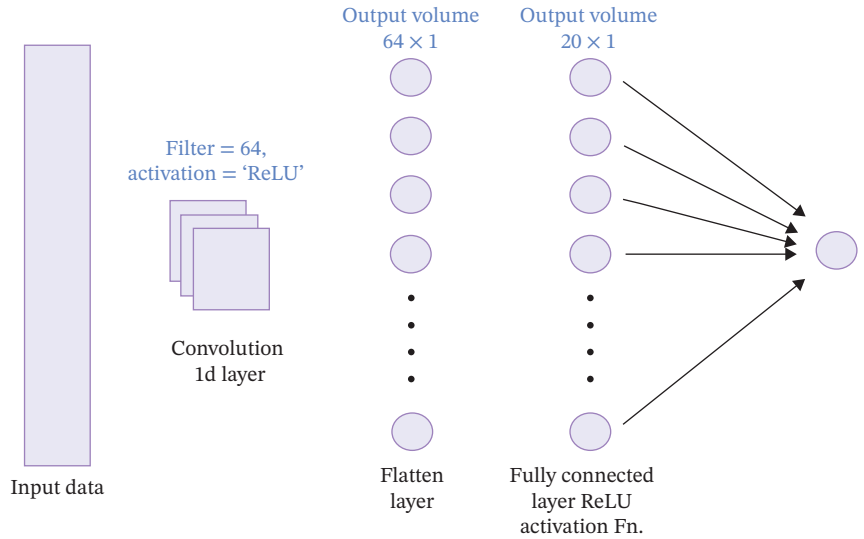

FIGURE 6 | Single convolutional neural network (CNN) structure.

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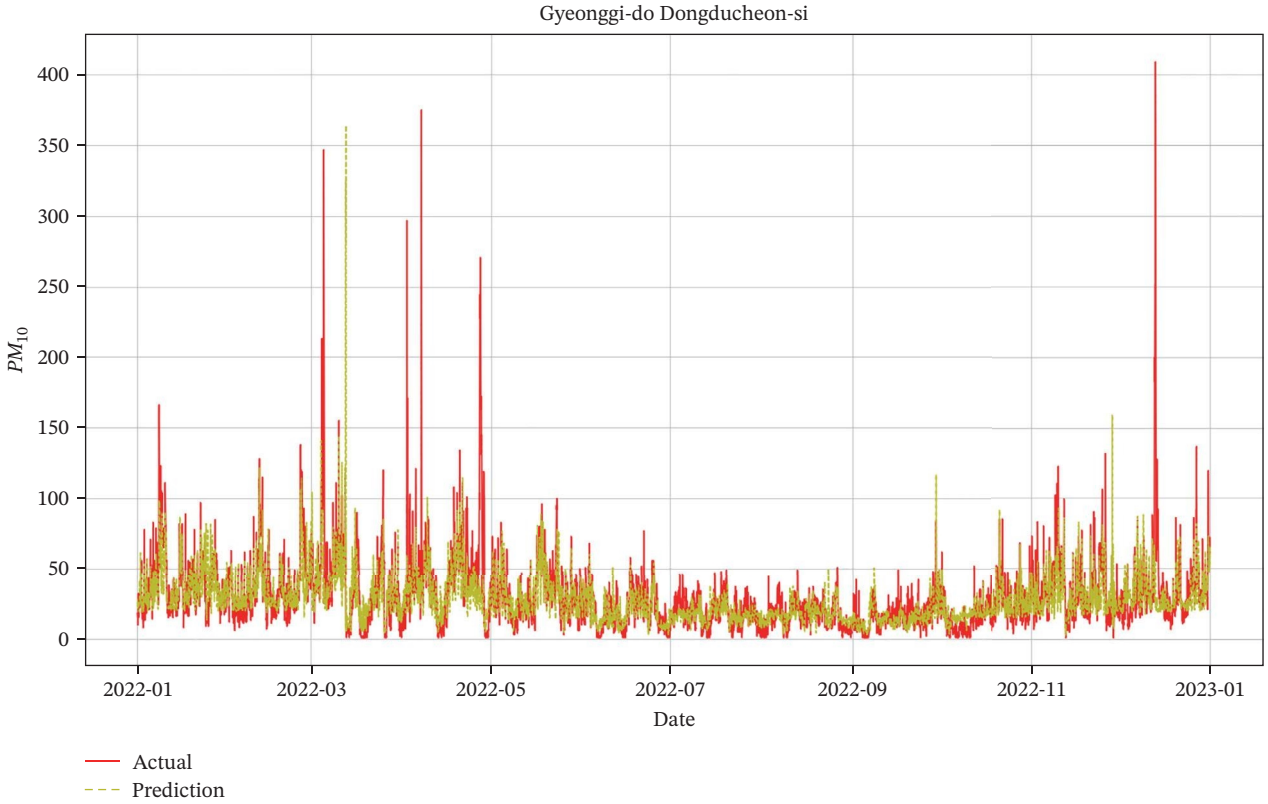

FIGURE 7 | Forecast graph of the convolutional neural network (CNN) bidirectional gated recurrent unit (BiGRU) model for Dongducheon-si.

improving predictive performance by comparing the MLP, basic ANN models, RNNs (RNN, LSTM, and GRU), and CNN, which are time-series DL models, and implementing a new hybrid combination model. The model showed improved MAE performance overall. However, in some cities, the RMSE was higher than that of simpler models such as MLP. This is likely due to the model's limitations in accurately capturing extreme fluctuations in PM10 concentrations, which tend to contribute more to RMSE. Future studies should expand this study by considering new variables that affect the prediction of the fine-dust concentration and improving the prediction performance using clustering methods that group and predict regional characteristics. The proposed future research can improve the future fine-dust concentration prediction system, contributing to establishing future air pollution prevention plans. In this paper, although limited to three urban areas, the model can be extended to rural regions and additional cities for broader applicability. And in addition, in the future, it will be necessary to conduct research to further improve model interpretability by applying the SHAP methodology, an AI technique.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data used in this study are publicly available from the Korea Environment Corporation (AirKorea, https://www.airkorea.or.kr) and the Korea Meteorological Administration (https://data.kma.go.kr). All datasets are open access and can be freely obtained from the corresponding websites.

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